CARDIFF UNIVERSITY
EXAMINATION PAPER

Academic Year :	2019-2020
Examination Period :	Spring
Examination Paper Number :	MA 0367
Examination Paper Title :	Time Series Analysis and Forecasting

Do not turn this page over until instructed to do by the Senior Invigilator.

Structure of Examination Paper:

There are Five pages.
There are FOUR questions in total.
There are no appendices.
The mark obtainable for each question is 25.

Students to be provided with:

The following items of stationery are to be provided at the start of the examination:
One answer book.

Instructions to Students:

Answer **THREE** questions

The use of a translation dictionary between English or Welsh and another language, provided that it bears an appropriate departmental stamp, is permitted in this examination.

Question 1

(i) State the Bochner-Khinchin theorem for continuous and discrete stationary processes.

[2]

(ii) Write the definition of Brownian motion and derive it's mean function and covariance function.

[3]

(iii) Write R code for computing the h-step ahead forecast of the time series

$$3, -1, -2, 2, -1, 4$$

from an AR(4) process given by the difference equation with coefficients $a_1 = 0.8$, $a_2 = -0.6$, $a_3 = 0.35$, $a_4 = 0.16$, mean $\mu = 0$ and $h = 1, 2, \ldots, 100$.

[3]

(iv) Consider the time series $x_k = 3\sin(11\pi k/21) + 2\sin(8\pi k/17) + \sin(9\pi k/14)$, $k \in \mathbb{Z}$, and the filter with spectrum $f(\lambda) = 1_{[0.5\pi, 0.6\pi]}(\lambda)$, $\lambda \in (0, \pi)$. Find the filtered series and its spectrum.

[2]

(v) Consider the time series $x_1 = 33$, $x_2 = 37$, $x_3 = 38$ and assume that the observations correspond to a stationary normal process with covariance function $R(k) = 1.8e^{-|k|}$. Compute the best linear predictor for x_5 and also the prediction bounds for x_5 with confidence level $1 - \alpha$.

[6]

(vi) Suppose that the time series $3, -2, 1, -2, -3, 1$ is a realization from an AR(2) process $x_t - a_1 x_{t-1} - a_2 x_{t-2} = \epsilon_t$, where ϵ_t is white noise. Find estimates of the parameters of the AR(2) process, that is, the coefficients a_1, a_2, and the noise variance $\sigma^2 = \mathrm{Var}(\epsilon_t)$.

[5]

(vii) Find values of the parameter c such that the stochastic process x_t satisfying

$$x_t = 2x_{t-1} - 4cx_{t-3} + 8cx_{t-4} + \varepsilon_t, \quad t = 0, \pm 1, \ldots,$$

is stationary and values of c when x_t is non-stationary, where ϵ_t is a white noise process with $\mathrm{E}[\epsilon_t] = 0$ and $\mathrm{Var}\,[\epsilon_t] = \sigma_\epsilon^2$.

[4]

Question 2

(i) State the Mercer expansion and the Karhunen-Loeve expansion.

[2]

(ii) Derive the formulae for the linear h-step ahead forecast of a causal ARMA process.

[4]

(iii) Write R code for computing the periodogram of a realization of length 150 from the white noise process with mean 2.3 and variance 9 without use of the Fourier transform.

[4]

(iv) Is the covariance function $R(k) = \frac{3}{(4+|k|^{0.75})^{1.4}}$, $k \in \mathbb{Z}$, short-range or long-range dependent? Explain your answer.

[2]

(v) Consider the causal AR(3) process $x_t = x_{t-1} - 0.6x_{t-2} + 0.4x_{t-3} + \epsilon_t$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Find the coefficients c_0, c_1, c_2, c_3, c_4 of the MA representation $x_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j}$ of this AR(3) process.

[5]

(vi) Suppose that the time series $4, 5, 8, 2, 4, 7$ is a realization from an invertible MA(1) process satisfying the difference equation $x_t = \mu + \epsilon_t + b_1\epsilon_{t-1}$, where ϵ_t is white noise. Estimate the parameters of this process: the mean μ, the coefficient b_1 and the noise variance $\sigma^2 = \mathrm{Var}(\epsilon_t)$.

[4]

(vii) Find the covariance function for a weakly stationary process x_t in discrete time and spectral density $f(\lambda) = 3 + \cos(\lambda)$.

[4]

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Question 3

(i) Give the definition of an invertible ARMA(p, q) process.

[2]

(ii) Consider a time series from a stationary (in wide sense) process with the spectral density $f(\lambda)$ which is such that $f(0) \neq 0$ and $f(0) < \infty$. Show that the average $\bar{X}_n$ of the time series satisfies $n\text{Var}(\bar{X}_n) \to 2\pi f(0)$.

[3]

(iii) Write R code for simulating a realization of a Gaussian process with the mean function $m(t) = 2/(3 + t^2)$ and the covariance function $R(t, s) = e^{-3|t-s|^{1.5}}$ at points $0.4, 0.7, 0.9, 1.4$.

[3]

(iv) Find a difference equation of the ARMA(p, q) model for which the correlation function $\rho(k)$ is such that $\rho(1) = 1/2$, $\rho(2) = 1/3$ and $\rho(k) = 0$ for $k = 3, 4, \ldots$.

[5]

(v) Let a process x_t satisfy $x_t - 1.2x_{t-1} + 0.3x_{t-2} = \epsilon_t + 0.8\epsilon_{t-1} + 0.7\epsilon_{t-2}$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Identify if x_t is invertible or not. Give a reason for your answer.

[2]

(vi) A company produces and sells hearing devices. Sales of devices (in units of 100) over the past 12 weeks are $x_1, \ldots, x_{12}$ with $x_{12} = 4.1$. The company believes that the AR(1) process $x_t - 3 = 0.9(x_{t-1} - 3) + \epsilon_t$, $t \in \mathbb{Z}$, where ϵ_t is Gaussian white noise with zero mean and variance 1.96, is relevant for forecasting. Forecast sales for weeks 13,14,15 and 16, and for each of the 4 weeks give an expression for a confidence interval for the forecast with confidence level $1 - \alpha$.

[4]

(vii) Consider the stochastic process x_t defined by

$$x_t = 1 + \xi \cos(\lambda t + \eta),\ t = 0, \pm 1, \ldots,$$

where ξ is a random variable with zero mean and unit variance, η is a random variable with a uniform distribution on the interval $[-\pi, \pi]$ independent of ξ, and λ is a positive number. Show that this process is stationary in the wide sense. Find the covariance function of the process x_t.

[6]

Question 4

(i) Give the definition of an n-order stationary process and the definition of the Wiener process.

[2]

(ii) State and prove the condition of causality of the AR(2) process in terms of coefficients.

[5]

(iii) Write R code for finding values of the correlation function $\rho(1), \rho(2), \rho(3), \rho(4), \rho(5)$ for an AR(3) process satisfying $x_t = 0.8x_{t-1} - 0.5x_{t-2} + 0.15x_{t-2} + \varepsilon_t$, where ϵ_t is white noise, $t \in \mathbb{Z}$.

[3]

(iv) Let a process x_t satisfy $x_t + 0.4x_{t-1} + 0.9x_{t-2} = \epsilon_t - 1.4\epsilon_{t-1}$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Identify if x_t is causal or not. Give a reason for your answer.

[2]

(v) Sketch the temporal and spectral representations of realizations of two stochastic processes $x_k = 2\sin(2\pi k/11) + \epsilon_k$ and $y_k = \sin(2\pi k/22) + 3\varepsilon_k$, where ϵ_k and ε_k are white noise with mean 0 and variance 1. Explain any differences in your sketches.

[3]

(vi) Suppose that the average precipitation in Dublin in successive months was approximated by the AR(2) process $x_t - 15 = 0.8(x_{t-1} - 15) - 0.4(x_{t-2} - 15) + \epsilon_t$, where ϵ_t is Gaussian white noise with zero mean and variance 0.9. Suppose that the last successive observations are 13, 16, 18, 14. Find predictions for the next two months, the mean square error of predictions, and the confidence intervals with confidence level $1 - \alpha$.

[5]

(vii) Find the variance and correlation function of the AR(2) process $x_t = 0.8x_{t-1} - 0.3x_{t-2} + \epsilon_t$, $t \in \mathbb{Z}$, where ϵ_t is white noise with zero mean and variance $\sigma^2 = 1.8$, using Yule-Walker equations.

[5]